

Mathieu Celerier

POSTDOCTORAL RESEARCHER

✉ mathieu.celerier.ing@gmail.com | 📅 1999/08/09 | 🌐 mathieu-celerier | 📄 mathieu-celerier | 📞 0000-0001-8290-3577 | 🏠 Mathieu Celerier

Summary

Postdoctoral researcher at the CNRS-AIST Joint Robotics Laboratory in Tsukuba, Japan. I study how robots generate motion, respond safely to physical contact, and learn collaborative behavior from interaction.

My PhD developed online disturbance-aware motion generation and an explicit-compliance framework for torque-controlled robots. My current work connects physical human-robot interaction with imitation learning for collaborative manipulation.

Work Experience

CNRS-AIST Joint Robotics Laboratory

POSTDOCTORAL RESEARCHER

Tsukuba, Japan

February 2026 - Present

Conduct research on understanding and improving physical human-robot interaction.

Informatic, Robotic and Microelectronic Laboratory of Montpellier (LIRMM)

Montpellier, France

ROBOTICS INTERNSHIP

Feb. 2021 - July 2021

- Assisted PhD student on the subject of dual arm manipulation of deformable objects.
- Development in C++ of robotic applications using the Robot Kinematic Control Library (RKCL) for dual arm manipulation of deformable objects.
- Assisted the PhD student with Python programming for optimization/improvement and for linking Python and C++.

Informatic, Robotic and Microelectronic Laboratory of Montpellier (LIRMM)

Montpellier, France

ROBOTICS INTERNSHIP

May 2020 - Aug. 2020

- V-REP (CoppeliaSim) simulation model of HOAP-3 robot.
- Development in C++ of robotic applications using the Robot Kinematic Control Library (RKCL) for dual arm manipulation of deformable objects.
- Assisted the PhD student with Python programming for optimization/improvement and for linking Python and C++.

Bristol Robotics Laboratory

Bristol, United Kingdom

UNDERGRADUATE INTERNSHIP

June 2019 - Aug. 2019

- Development of a web application implementing scratch programming framework (Blockly) for Anki's Vector robot. The target was to help children discover robotics and programming for the Robotics Week in UK.
- Development of a ROS1 wrapper for Anki's Vector robot API, and implementation of the ROS Navigation Stack on the robot.

Education

University of Montpellier CNRS-AIST Joint Robotics Laboratory

Tsukuba, Japan

PH.D

Sep. 2022 - Jan. 2026

Conduct research on understanding and improving physical human-robot interaction.

University of Quebec in Chicoutimi

Remote from France due to

COVID-19

STUDY SEMESTER

Sep. 2020 - Dec. 2020

Courses on Neural Networks, Quantitative methods for engineers, Engineering project management and Advance numeric communication systems.

Polytech Montpellier

Montpellier, France

MASTER DEGREE IN ENGINEERING

Sep. 2018 - Aug. 2021

- Microelectronics and Automation Major, Specialized in Robotics.
- Smart Bird Feeder project: C programming on STM32, Low-Power constraints, Logging system, Picture capture when a bird arrives and saving on SD card.
- Gravity torque computing for compliance project: Developed Python URDF parser, recursive tree exploration, gravity torque computing with Python/C++.
- Humanoid walk study project: Center of mass trajectory modeling from each step balance point under MATLAB.
- 6 DoF manipulator arm project: Modelling of a manipulator arm using modified Denavit-Hartenberg parameters, dynamic control of the arm, made from scratch using Julia programming language.

IUT of Nîmes

Nîmes, France

TECHNICAL ASSOCIATE'S DEGREE IN INDUSTRIAL COMPUTING AND ELECTRICAL ENGINEERING

Sep. 2016 - Aug. 2018

- Machine learning project to recognize poker cards:
- C++ programming.
- Neural Network based recognition.
- Neural Network algorithms (MLP/CNN) self-taught, made from scratch (Python/MATLAB).